CareTrace: A Neurosymbolic Approach to Pediatric After-Hours Triage

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ABSTRACT

After-hours pediatric triage is a high-stakes setting in which caregivers seek guidance for acutely ill children without immediate access to clinicians. Large language models (LLMs) are increasingly capable of producing plausible clinical language, but pure generative approaches lack verifiable reasoning provenance, cannot guarantee adherence to established clinical practice guidelines (CPGs), and risk fabricating medication dosing or epidemic context. **CareTrace** is a neurosymbolic prototype that grounds pediatric triage in the Seattle Children's Fever CPG by combining (1) an LLM or rule-based natural language interpretation layer, (2) a SNOMED CT–backed knowledge graph in Neo4j AuraDB populated from the CPG's red-flag criteria, (3) deterministic PyDatalog triage rules with full audit trace, and (4) a provenance-annotated explanation agent. A LangGraph orchestration pipeline sequences these components into a multi-turn conversational system. On three manually crafted scenarios spanning home management, emergency escalation, and urgent same-day care, CareTrace produces correct dispositions with full symbolic justification, while a vanilla LLM baseline cannot guarantee structured CPG adherence. The architecture demonstrates how neurosymbolic integration can improve safety, transparency, and auditability in conversational clinical decision support.

1. INTRODUCTION

1.1 Motivation

Pediatric fever is one of the most common reasons families contact medical advice lines after hours. Guidelines such as the Seattle Children's Fever CPG define clear escalation criteria — seizure, altered consciousness, breathing difficulty, extreme fever, infant under three months — that must be applied consistently. Parents describing symptoms in natural language often omit details critical for safe triage, and any system that fails to elicit those details risks either over-escalating (unnecessary ER visits) or under-escalating (missed emergencies).

Pure LLMs present two risks in this setting: (a) they cannot guarantee that CPG thresholds are honored because reasoning is implicit in weights rather than explicitly asserted, and (b) they can hallucinate — generating plausible-sounding but incorrect medication dosing, epidemiological context, or age-specific contraindications. Prior work in clinical NLP has shown that symbolic reasoning layers, when coupled with neural language understanding, yield more auditable and safer outputs in structured medical domains.

CareTrace was designed to demonstrate this neurosymbolic integration at prototype scale, with a deliberately bounded clinical scope: febrile illness, gastrointestinal symptoms, and dehydration risk in pediatric patients.

1.2 Contributions

1. **Neurosymbolic triage pipeline:** LangGraph state machine sequencing LLM interpretation, SNOMED-grounded KG retrieval, PyDatalog symbolic rules, and templated explanation with provenance strings.
2. **CPG-derived mini knowledge graph:** SNOMED CT concepts seeded from the Seattle Children's Fever CPG stored in Neo4j AuraDB with `IS_A` hierarchy and attribute edges.
3. **Audit-traceable reasoning:** every disposition includes a named `rule_ids` set explaining exactly which symbolic predicate fired.
4. **Medication safety hooks:** explicit CPG citations for acetaminophen and ibuprofen age gates with provenance strings tying every statement to the source document URL.
5. **Comparative baseline:** a vanilla LLM baseline without symbolic gates enabling direct evaluation of provenance and consistency differences.

2. BACKGROUND

2.1 Neurosymbolic AI

Neurosymbolic AI combines the pattern-recognition strengths of neural networks with the deductive reasoning guarantees of symbolic systems. In the clinical domain this combination is particularly valuable: neural components handle the variability and ambiguity of natural language, while symbolic components enforce hard constraints derived from domain knowledge. More recently, LLM-plus-symbolic frameworks have emerged for structured document question answering and clinical guideline compliance.

2.2 Clinical Practice Guidelines and Pediatric Triage

Clinical practice guidelines are evidence-based documents that define care protocols for defined conditions. The Seattle Children's Fever CPG specifies fever thresholds, escalation triggers, and medication guidance aligned with the American Academy of Pediatrics. Encoding CPG logic as machine-executable predicates is an established approach in clinical decision support systems (CDSS). CareTrace applies this approach in a conversational setting where intake fields must be elicited dynamically from unstructured caregiver language.

2.3 Knowledge Graphs for Medical Concept Grounding

SNOMED CT is the world's most comprehensive clinical terminology with over 350,000 active concepts hierarchically organized through `IS_A` and attribute relationships. Snowstorm is an open-source SNOMED CT terminology server that exposes FHIR-compliant REST APIs. Neo4j is a native property graph database widely used in biomedical knowledge graph research. Combining Snowstorm concept retrieval with Neo4j graph persistence enables compact, query-friendly SNOMED subgraphs that can ground natural language terms to standard clinical identifiers.

3. SYSTEM ARCHITECTURE

3.1 Overview

CareTrace is implemented as a Python package (`caretrace`) with the following principal layers:

```
caretrace/
  main.py                ← CLI entrypoint
  config.py              ← Settings / .env loading
  state.py               ← CareTraceState, CaseFields,
                           TriageDecision
  orchestration/graph.py← LangGraph pipeline
  agents/interpretation ← Text → CaseFields
  agents/explanation    ← Decision → reply
  agents/medication     ← CPG medication guidance
  logic/triage_rules.py ← PyDatalog rules
  graph/snomed_retrieval← Neo4j SNOMED lookup
  graph/fever_cpg_mentions← CaseFields → KG terms
  graph/neo4j_client    ← Driver + run_cypher
  evaluation/harness    ← Scenario replay
KG_implementation/
  Scaffold_KG_FeverCPG_Text_to_AuraDB.ipynb
  Pediatric_Fever_KG_Snowstorm_to_AuraDB.ipynb
```

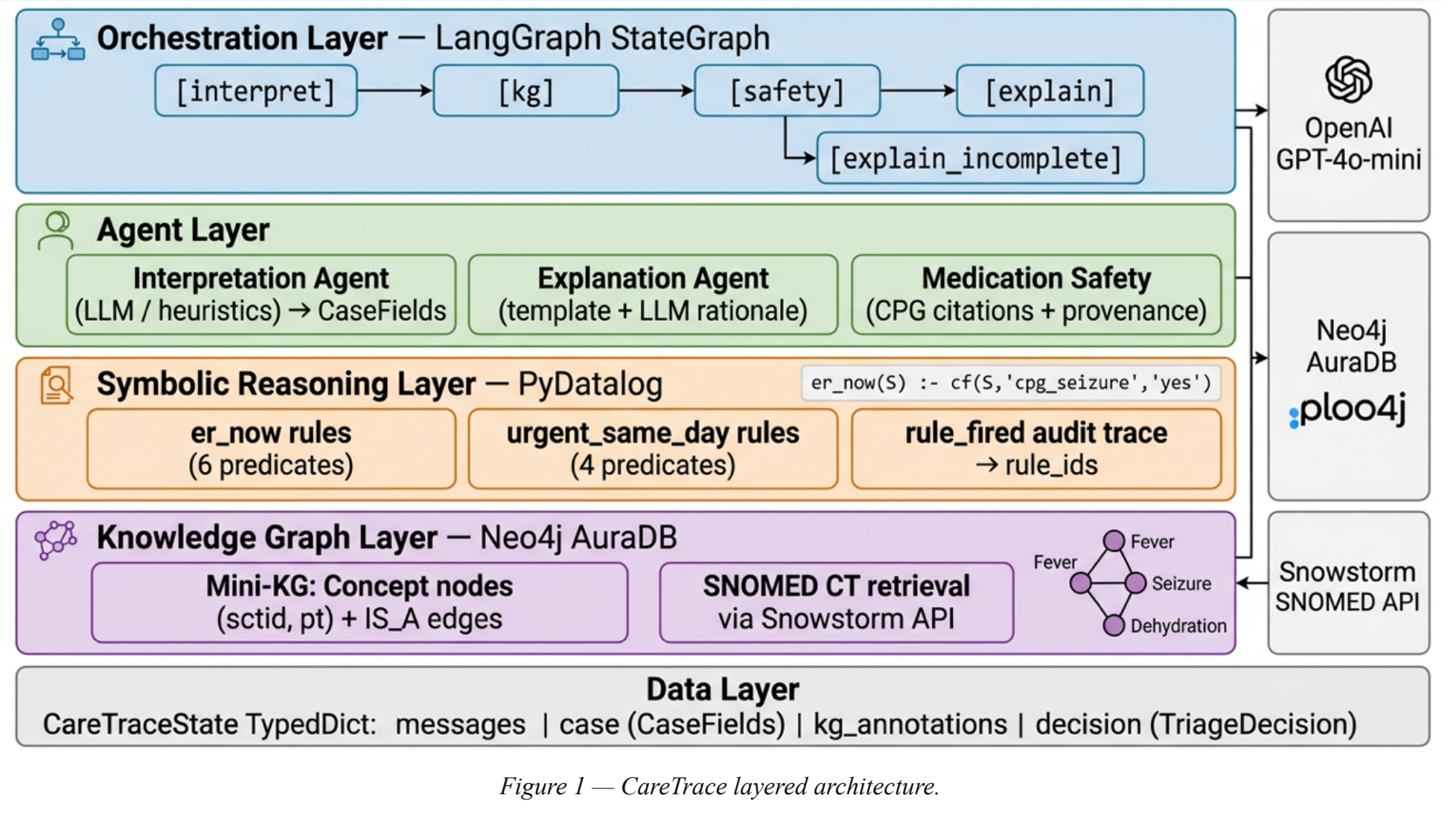

CareTrace — Neurosymbolic Pediatric Triage System
Orchestration Layer — LangGraph StateGraph
[interpret]
[kg]
[safety]
[explain]
[explain_incomplete]
OpenAI GPT-4o-mini
Agent Layer
Interpretation Agent (LLM / heuristics) → CaseFields
Explanation Agent (template + LLM rationale)
Medication Safety (CPG citations + provenance)
Neo4j AuraDB
neo4j
Symbolic Reasoning Layer — PyDatalog
er_now(S) :- cf(S,'cpg_seizure','yes')
er_now rules (6 predicates)
urgent_same_day rules (4 predicates)
rule_fired audit trace → rule_ids
Knowledge Graph Layer — Neo4j AuraDB
Mini-KG: Concept nodes (sctid, pt) + IS_A edges
SNOMED CT retrieval via Snowstorm API
Fever
Fever
Seizure
Dehydration
Snowstorm SNOMED API
Data Layer
CareTraceState TypedDict: messages | case (CaseFields) | kg_annotations | decision (TriageDecision)

Figure 1 — CareTrace layered architecture.

3.2 State Model

All per-session data flows through a single `CareTraceState` TypedDict shared across every LangGraph node. The two most important nested types are `CaseFields` and `TriageDecision`.

Table 1 — CareTraceState pipeline fields

Field	Type	Purpose
`messages`	`list[dict]`	Full chat history (role + content)
`raw_user_text`	`str`	Latest caregiver utterance
`case`	`CaseFields`	Canonical clinical fields, accumulated across turns
`kg_annotations`	`list[dict]`	SNOMED concept + ancestor rows from Neo4j
`decision`	`TriageDecision`	Disposition, rule_ids, med_flags, missing_required
`assistant_reply`	`str`	Final reply rendered to caregiver
`turn`	`int`	Turn counter (incremented per invoke)

`CaseFields` is a TypedDict with 11 clinical fields. Each field is independently extractable and independently defaults to `None` / `"unknown"` so partial turns are safe. The full schema is:

Table 2 — CaseFields schema (all fields)

Field	Type	Allowed Values
`temp_f`	`float | None`	Numeric °F; `None` if not stated
`temp_unknown`	`bool`	`True` if caregiver explicitly says temperature is unknown
`age_years`	`float | None`	Age in years
`age_months`	`float | None`	Age in months (takes precedence for infant gates)
`weight_kg`	`float | None`	Weight in kg; required for dose calculation
`alertness`	`str`	`normal | sleepy_ok | altered | unknown`
`breathing`	`str`	`normal | tachypnea_concern | distress | unknown`
`fluid_intake`	`str`	`good | some | poor | none | unknown`
`urine_last_8h`	`str`	`yes | no | unknown`
`vomiting`	`str`	`none | once | repeated | unknown`
`seizure`	`str`	`yes | no | unknown`
`fever_duration_hours`	`float | None`	Duration in hours; ≥72 triggers 3-day CPG rule
`current_meds`	`list[str]`	Medication names (e.g. `["amoxicillin"]`)
`local_outbreak_context`	`str | None`	Community illness context — does NOT modify rules

`TriageDecision` is the structured output of the safety node:

```
@dataclass
class TriageDecision:
  disposition: str        # ER_NOW | URGENT_SAME_DAY
                          # | HOME_MANAGEMENT | OUT_OF_SCOPE
  rule_ids: list[str]     # e.g. ["R_CPG_SEIZURE"]
  missing_required: list[str]  # [] when intake complete
  med_flags: list[str]    # e.g. ["cpg_ibuprofen_under_6mo"]
  out_of_scope_reason: str | None
```

Multi-turn accumulation — `_merge_non_empty`: Each turn extracts a *delta* dict from the current utterance. Fields with value `None`, empty string, empty list, or `"unknown"` (for categorical fields) are excluded from the merge. This ensures that a caregiver omitting temperature in turn 2 does not overwrite the temperature reported in turn 1:

```
def _merge_non_empty(base: CaseFields,
                     delta: dict) -> CaseFields:
    out = dict(base)
    for k, v in delta.items():
        if v is None: continue
        if isinstance(v, str) and v == "": continue
        if k in CATEGORICAL and v == "unknown": continue
        if isinstance(v, list) and len(v) == 0: continue
        out[k] = v   # overwrite only with real values
    return out
```

3.3 LangGraph Orchestration Pipeline

Each caregiver turn triggers a single invocation of the compiled `StateGraph`. The graph is built once at import time via `build_app()` and reused across turns:

```
def build_app():
    g = StateGraph(CareTraceState)
    g.add_node("interpret",          node_interpret)
    g.add_node("kg",                 node_kg)
    g.add_node("safety",             node_safety)
    g.add_node("explain",            node_explain)
    g.add_node("explain_incomplete", node_explain_incomplete)

    g.set_entry_point("interpret")
    g.add_edge("interpret", "kg")
    g.add_edge("kg",        "safety")
    g.add_conditional_edges(
        "safety",
        route_after_safety,          # routing function
        {"explain":            "explain",
         "explain_incomplete": "explain_incomplete"},
    )
    g.add_edge("explain",            END)
    g.add_edge("explain_incomplete", END)
    return g.compile()
```

The routing function `route_after_safety` inspects the `decision.missing_required` list produced by the safety node:

```
def route_after_safety(state) -> Literal[
        "explain", "explain_incomplete"]:
    miss = (state.get("decision") or {}) \
               .get("missing_required") or []
    return "explain_incomplete" if miss else "explain"
```

Node `interpret` calls `interpret_user_message(settings, prior_case, text)`. If `OPENAI_API_KEY` is set and `CARETRACE_MOCK_LLM` is unset, it invokes `ChatOpenAI.with_structured_output(ExtractedCase)` to extract a validated Pydantic model. Otherwise it falls back to `_heuristic_extract`. In both cases the result is merged into the prior `CaseFields` via `_merge_non_empty`.

Node `kg` skips entirely when `CARETRACE_SKIP_NEO4J=1` (returns `{"kg_annotations": []}`). When enabled, it calls `kg_mentions_from_case_and_text(case, raw_text)` to derive search terms, then calls `annotate_case_mentions(driver, mentions, database=...)` which runs `generalize_symptom_mention` per term — concept lookup + ancestor traversal.

Node `safety` is pure Python + PyDatalog — no I/O, no LLM. It calls `required_missing(case)` then `evaluate_triage(case, missing)`.

Node `explain` / `explain_incomplete` both call the same `explain(settings, case, decision, kg)` function. The function dispatches on `decision.disposition`: if `OUT_OF_SCOPE` with reason `incomplete_intake`, it renders a question list from `missing_required`; otherwise it renders a full disposition reply with rule trace, medication guidance, and safety netting.

3.4 Interpretation Agent — Dual-Mode Extraction

The interpretation agent implements two execution paths behind a single interface. The Pydantic schema `ExtractedCase` defines the contract between both paths:

```
class ExtractedCase(BaseModel):
    temp_f:      float | None = None
    temp_unknown: bool = False
    age_years:   float | None = None
    age_months:  float | None = None
    weight_kg:   float | None = None
    vomiting:    str = "unknown"  # none|once|repeated
    alertness:   str = "unknown"  # normal|sleepy_ok|altered
    breathing:   str = "unknown"  # normal|tachypnea|distress
    fluid_intake: str = "unknown" # good|some|poor|none
    urine_last_8h: str = "unknown"# yes|no
    current_meds: list[str] = []
    seizure:     str = "unknown"  # yes|no
    fever_duration_hours: float | None = None
```

LLM path — when `OPENAI_API_KEY` is available, LangChain's `ChatOpenAI.with_structured_output(ExtractedCase)` instructs the model to populate every field using function-calling. The system prompt explicitly instructs: *"Use unknown when not stated. Never invent vitals. Map qualitative phrases conservatively."*

Heuristic path — a deterministic regex/keyword extractor. Representative patterns:

```
# Temperature: "103.5 F", "101°F", "temp reading 102"
re.search(r"(\d{2,3}(?:\.\d+)?)\s*(?:°?\s*f|f\b)", t)

# Alertness — hard keywords
if any(x in t for x in ("barely responding",
       "won't wake","hard to wake","lethargic")):
    out["alertness"] = "altered"
elif any(x in t for x in ("tired but","answers","talks")):
    out["alertness"] = "sleepy_ok"

# Vomiting quantification
if re.search(r"\b([2-9]|\d{2,})\s*times\b", t):
    out["vomiting"] = "repeated"
elif any(x in t for x in ("once","one time")):
    out["vomiting"] = "once"
elif "vomit" in t or "threw up" in t:
    out["vomiting"] = "once"   # unquantified → conservative

# Fever duration → CPG 3-day rule
m = re.search(r"fever[^\n]{0,40}(\d{1,2})\s*days?", t)
if m: out["fever_duration_hours"] = float(m.group(1))*24
```

3.5 Symbolic Triage Rules — PyDatalog

Rules are defined once at module import using PyDatalog's Python DSL. Each fact is scoped to a *per-evaluation UUID session* (`cf(session_id, name, value)`) so that parallel evaluation runs do not share state. The full rule set:

ER hard gates (highest priority):

```
er_now(S) <= cf(S, "alertness",          "altered")
er_now(S) <= cf(S, "breathing",          "distress")
er_now(S) <= cf(S, "dehydration_severe", "yes")
er_now(S) <= cf(S, "fluid_intake", "none") \
           & cf(S, "urine_last_8h", "no")
er_now(S) <= cf(S, "cpg_seizure",        "yes")
er_now(S) <= cf(S, "cpg_infant_under_3mo_fever", "yes")
```

Urgent same-day patterns:

```
urgent_same_day(S) <= cf(S,"temp_f","very_high") \
                    & ~er_now(S)
urgent_same_day(S) <= cf(S,"vomiting","repeated") \
                    & cf(S,"fluid_intake","poor") \
                    & ~er_now(S)
urgent_same_day(S) <= cf(S,"breathing","tachypnea_concern") \
                    & ~er_now(S)
urgent_same_day(S) <= cf(S,"cpg_fever_duration_3_days","yes") \
                    & ~er_now(S)
```

Home management candidate:

```
home_candidate(S) <= cf(S,"alertness","normal") \
    & cf(S,"breathing","normal") \
    & ~cf(S,"fluid_intake","none") \
    & ~er_now(S) & ~urgent_same_day(S)
```

The `_case_to_facts` function translates `CaseFields` into session-scoped facts before rule evaluation. Derived facts are computed here: temperature is bucketed (`very_high` ≥ 104 °F, `high` ≥ 103 °F), severe dehydration is asserted when both `fluid_intake ∈ {poor, none}` and `urine_last_8h = no`, and CPG predicates (`cpg_infant_under_3mo_fever`, `cpg_fever_duration_3_days`, `cpg_ibuprofen_contraindicated_age`) are derived from numeric age and temperature thresholds.

Every rule that contributes to a disposition has a parallel `rule_fired(S, "R_ID")` predicate with identical premises. After evaluation, `rule_fired` is queried to enumerate all fired rule IDs, which are returned as `TriageDecision.rule_ids`.

3.6 Knowledge Graph Construction

Two Jupyter notebooks build the Neo4j AuraDB mini-KG using a shared `build_mini_kg(seed_ids, max_concepts=45)` function:

1. Accept a list of seed SCTIDs (resolved from CPG text via Snowstorm `/MAIN/concepts?term=...`).
2. BFS over `IS_A` parents (Snowstorm `/MAIN/concepts/{id}/ancestors`).
3. For each concept, fetch up to `MAX_ATTR_NEIGHBORS=3` attribute relationship targets (Snowstorm `/MAIN/concepts/{id}/relationships?active=true`), filtered to exclude `IS_A` (typeId=116680003).
4. Stop when total node count reaches `MAX_TOTAL_CONCEPTS=45`.
5. Second pass: add `IS_A` parents of newly added attribute targets (ancestor closure).

Each node and edge is loaded to Neo4j via `MERGE` to ensure idempotency:

```
// Node load
MERGE (c:Concept {sctid: $sctid})
  ON CREATE SET c.pt = $pt, c.kg_source = $src
  ON MATCH  SET c.pt = $pt

// IS_A edge
MATCH (child:Concept {sctid: $child_id})
MATCH (parent:Concept {sctid: $parent_id})
MERGE (child)-[:IS_A]->(parent)

// Attribute edge
MATCH (src:Concept {sctid: $src_id})
MATCH (tgt:Concept {sctid: $tgt_id})
MERGE (src)-[:REL {typeId: $typeId,
                   typeTerm: $typeTerm}]->(tgt)
```

3.7 Schema-Adaptive SNOMED Retrieval

`snomed_retrieval.py` supports three different graph schemas that arise from different loading methods. To avoid Neo4j planner warnings (`01N50`, `01N52`) for non-existent labels or properties, the retrieval layer probes the graph schema at first use and caches results per driver instance:

```
CALL db.labels()        YIELD label        → frozenset[str]
CALL db.propertyKeys()  YIELD propertyKey  → frozenset[str]
CALL db.relationshipTypes() YIELD ...      → frozenset[str]
```

The dispatch logic for concept lookup is:

```
def find_concepts_by_term(driver, term, limit=5):
    labels = _cached_labels(driver)
    keys   = _cached_property_keys(driver)

    # 1. CPG mini-KG: pt + sctid/conceptId
    if "pt" in keys and ("sctid" in keys
                         or "conceptId" in keys):
        hits = _find_concepts_pt_slice(...)
        if hits or "Description" not in labels:
            return hits   # commit to this schema

    # 2. Snowstorm RF2 loader: Description label
    if "Description" in labels:
        hits = _find_concepts_snowstorm(...)
        if hits: return hits

    # 3. Legacy / curated slice: term field
    if "term" in keys:
        return _find_concepts_legacy(...)
    return []
```

The mini-KG concept search query (schema 1):

```
MATCH (c:Concept)
WHERE c.pt IS NOT NULL
  AND (c.sctid IS NOT NULL OR c.conceptId IS NOT NULL)
  AND toLower(toString(c.pt)) CONTAINS toLower($term)
RETURN coalesce(toString(c.sctid),
                toString(c.conceptId)) AS id,
       toString(c.pt) AS term
LIMIT $limit
```

The ancestor traversal uses variable-depth `IS_A*1..8` and is likewise guarded by schema detection. The `RETURN` clause is dynamically built from whichever of `sctid`, `conceptId`, `id` are present in `db.propertyKeys()`:

```
MATCH (c:Concept)-[:IS_A*1..8]->(a:Concept)
WHERE {dynamic_id_match_clause}
RETURN DISTINCT
  coalesce(toString(a.sctid),
           toString(a.conceptId)) AS id,
  toString(a.pt) AS term
LIMIT 40
```

3.8 Medication Safety Module

`medication.py` provides CPG-grounded `MedGuidance` dataclass objects:

```
@dataclass(frozen=True)
class MedGuidance:
    text: str
    provenance: str   # always includes source URL
    must_escalate: bool
```

Decision logic for the two modeled antipyretics:

- **Acetaminophen:** `must_escalate=True` if age < 3 months; if `weight_kg` is `None`, refuse to compute a dose and request weight; otherwise return illustrative 15 mg/kg with a verification disclaimer.
- **Ibuprofen:** `must_escalate=True` if age < 6 months; if dehydration risk is present, advise against ibuprofen and prefer acetaminophen.

PyDatalog `med_flag` predicates mirror the same age gates in the rule trace (`cpg_ibuprofen_contraindicated_age`, `cpg_no_routine_antipyretic_under_3mo`, `antibiotic_on_file_review_interactions`, `dehydration_avoid_nsaid`), so medication constraints appear in both the structured `TriageDecision.med_flags` output and the natural language explanation.

3.9 AGENT DESIGN

CareTrace contains three specialized agents, each with a distinct responsibility and execution mode. Their positions in the pipeline are summarized in Table 9.

Table 9 — Agent overview

Agent	Module	Input	Output	Modes
Interpretation	`agents/interpretation.py`	Raw caregiver text + prior `CaseFields`	Updated `CaseFields`	LLM structured extraction / Heuristic regex
Medication Safety	`agents/medication.py`	`CaseFields`	`MedGuidance` objects	Deterministic (rule-based only)
Explanation	`agents/explanation.py`	`CaseFields` + `TriageDecision` + `kg_annotations`	Natural language reply string	Template / LLM / LAG (Logic-Augmented Generation)

3.9.1 Interpretation Agent

The interpretation agent is the only agent that runs *before* the symbolic layer. It bridges unstructured natural language and the typed `CaseFields` schema. Its dual-mode design (described in §3.4) ensures the system works offline with deterministic heuristics and in production with LLM-based structured output.

A key design constraint is *conservatism*: the heuristic path never infers a value from absence of evidence. For example, if breathing is not mentioned in an utterance, it remains `"unknown"` rather than defaulting to `"normal"`. This forces the required intake policy to prompt the caregiver, preventing silent assumptions from bypassing safety gates.

The LLM path enforces the same conservatism via the system prompt: *"Use unknown when not stated. Never invent vitals. Map qualitative phrases conservatively."* The structured output contract (`ExtractedCase` Pydantic model) guarantees that the LLM cannot return fields outside the defined vocabularies — function-calling validates the schema before the response reaches the pipeline.

3.9.2 Medication Safety Agent

The medication safety agent is entirely deterministic — it contains no LLM call and no heuristics. Its sole responsibility is to apply CPG-grounded age and dehydration gates for acetaminophen and ibuprofen, returning `MedGuidance` objects with explicit provenance strings.

The design principle is *provenance completeness*: every piece of medication guidance the caregiver sees is accompanied by a citation string pointing to the exact CPG source URL. This makes the medication layer independently auditable — a reviewer can verify each statement against the original CPG without understanding the rest of the system.

The agent is also the only agent that explicitly **refuses to compute**: when `weight_kg` is absent, it returns a guidance object instructing the caregiver to confirm weight rather than estimating from age. This directly addresses one of the documented LLM failure modes (§5.5 — speculative dose computation from partial information).

3.9.3 Explanation Agent — Three Execution Modes

The explanation agent is the most architecturally interesting component. It implements three execution modes, selected at runtime via environment variables:

Table 10 — Explanation agent execution modes

Mode	Trigger	Mechanism	Trade-off
Template	`CARETRACE_MOCK_LLM=1` or no API key	String templates with rule trace and CPG summary inserted	Fully deterministic; no naturalness
LLM	API key set, `CARETRACE_USE_LAG` unset	LLM receives `case` + `decision` dict; generates free-form reply guided by system prompt	Natural language; weaker rule binding
LAG	`CARETRACE_USE_LAG=1`	LLM receives complete symbolic reasoning chain as structured context; constrained to express, not reason	Best combination: natural + symbolic-grounded

Template mode — the fallback path — directly inserts `rule_ids`, medication guidance, and CPG summary text into a fixed reply structure. The output is fully reproducible: identical `CaseFields` always produce identical text. The four template branches are: `incomplete_intake` (question list), `ER_NOW`, `URGENT_SAME_DAY`, and `HOME_MANAGEMENT`.

LLM mode — passes `case` and `decision` as a raw dict to `ChatOpenAI` (temperature=0.2) with a system prompt that constrains output style but does not explicitly provide the rule evaluation trace. The model must infer its explanation from the structured input. This is the weakest mode in terms of provenance binding.

Logic-Augmented Generation (LAG) mode — the most novel aspect of the explanation agent. Instead of asking the LLM to *reason* about the case, it is given the complete symbolic reasoning chain as a structured context block and asked only to *express* it in natural language. The context block passed to the LLM includes:

```
=== SYMBOLIC TRIAGE CONTEXT ===

PATIENT STATE (structured from caregiver reports):
- Age: 6 years
- Temperature: 103.5°F  [classified: high (≥103°F)]
- Alertness: ALTERED — not responding normally
- Breathing: normal
- Fluid intake: NONE (not drinking)
- Urination last 8h: NO (dry for 8+ hours)
- Vomiting: once
- Current medications: none reported

KNOWLEDGE GRAPH EVIDENCE:
  - Fever [SNOMED 386661006] ← matched phrase: "fever"
  - Dehydration [SNOMED 34095006] ← matched: "not drinking"

TRIAGE RULES EVALUATED (all rules in scope):
✓ FIRED: R_ER_ALERTNESS
  Label: Altered alertness
  Condition: alertness == altered
  CPG basis: Seattle Children's CPG: child not alert
             when awake → ER immediately
✓ FIRED: R_ER_DEHYDRATION_SEVERE
  Condition: dehydration_severe == yes
             (poor/no fluid AND no urine)
✗ NOT FIRED: R_CPG_SEIZURE  (Febrile seizure)
✗ NOT FIRED: R_URGENT_VERY_HIGH_FEVER  (Very high fever)

FINAL SYMBOLIC DECISION: ER_NOW

=== YOUR TASK ===
Express the above symbolic reasoning as a clear,
trustworthy caregiver-facing explanation.
Constraints:
1. State the disposition first — do NOT change it.
2. Explain WHY using the specific rules that fired.
3. Acknowledge which red-flag rules were NOT fired.
4. List medication flags and caregiver actions.
5. Do NOT add clinical content not in the context.
6. Do NOT invent diagnoses or conditions.
```

This approach — providing both fired and *not-fired* rules — allows the LLM to generate explanations that acknowledge which red flags were explicitly evaluated and ruled out, improving caregiver trust. The eight hard constraints in the task section prevent the LLM from modifying the disposition, fabricating new clinical content, or computing unsupported doses.

The rule metadata dictionary (`_RULE_META`) maps every rule ID to a triple of (human label, condition description, CPG basis), ensuring the structured context always includes a plain-English and CPG-traceable description of each rule alongside its machine predicate:

Table 11 — Rule metadata used in LAG context block

Rule ID	Human Label	CPG Basis
`R_ER_ALERTNESS`	Altered alertness	Child not alert when awake → ER immediately
`R_ER_BREATHING`	Breathing distress	Trouble breathing → ER immediately
`R_ER_DEHYDRATION_SEVERE`	Severe dehydration	No urine 8h + poor intake → ER immediately
`R_CPG_SEIZURE`	Febrile seizure	Febrile convulsion → ER immediately
`R_CPG_INFANT_UNDER_3MO_FEVER`	Infant <3 months with fever	Fever in infant <3 months → ER immediately
`R_URGENT_VERY_HIGH_FEVER`	Very high fever ≥104°F	Fever ≥104°F → same-day urgent evaluation
`R_URGENT_REPEATED_VOMIT_POOR_FLUID`	Repeated vomiting + poor intake	Vomits often + poor intake → same-day urgent
`R_URGENT_FEVER_OVER_3_DAYS`	Fever >3 days	Fever >3 days → same-day urgent evaluation
`R_HOME_CONSERVATIVE`	Conservative home management	Alert, breathing normally, taking fluids — no red flags

All three modes fall back gracefully: if the LLM call raises any exception (network failure, API quota, etc.), the agent silently falls back to the template path. This ensures the system always produces a structured, safe reply even without LLM connectivity.

4. KNOWLEDGE GRAPH DESIGN

4.1 Seed Concepts from CPG Extraction

Table 8 — CPG-seeded SNOMED CT anchor concepts

Query	SCTID	Preferred Term
fever	386661006	Fever
seizure	91175000	Seizure
dyspnea	267036007	Dyspnea
tachypnea	271823003	Tachypnea
lethargy	214264003	Lethargy
oliguria	83128009	Oliguria
dehydration	34095006	Dehydration
vomiting	422400008	Vomiting

4.2 Bounded Graph Expansion

The `build_mini_kg` function performs breadth-first expansion up to `MAX_TOTAL_CONCEPTS = 45`, following `IS_A` parents and up to `MAX_ATTR_NEIGHBORS = 3` attribute edges per concept for up to `MAX_HOPS_ATTR = 1` hops. A second ancestor-closure pass adds any remaining `IS_A` parents without expanding attribute edges, ensuring a semantically coherent subgraph without unbounded growth.

4.3 Runtime Retrieval

At each conversational turn the KG node calls `kg_mentions_from_case_and_text` to produce search terms (combining structured case signals with substring cues from the utterance), queries Neo4j with case-insensitive `CONTAINS` on `pt`, and traverses `IS_A` edges up to 8 hops for semantic context.

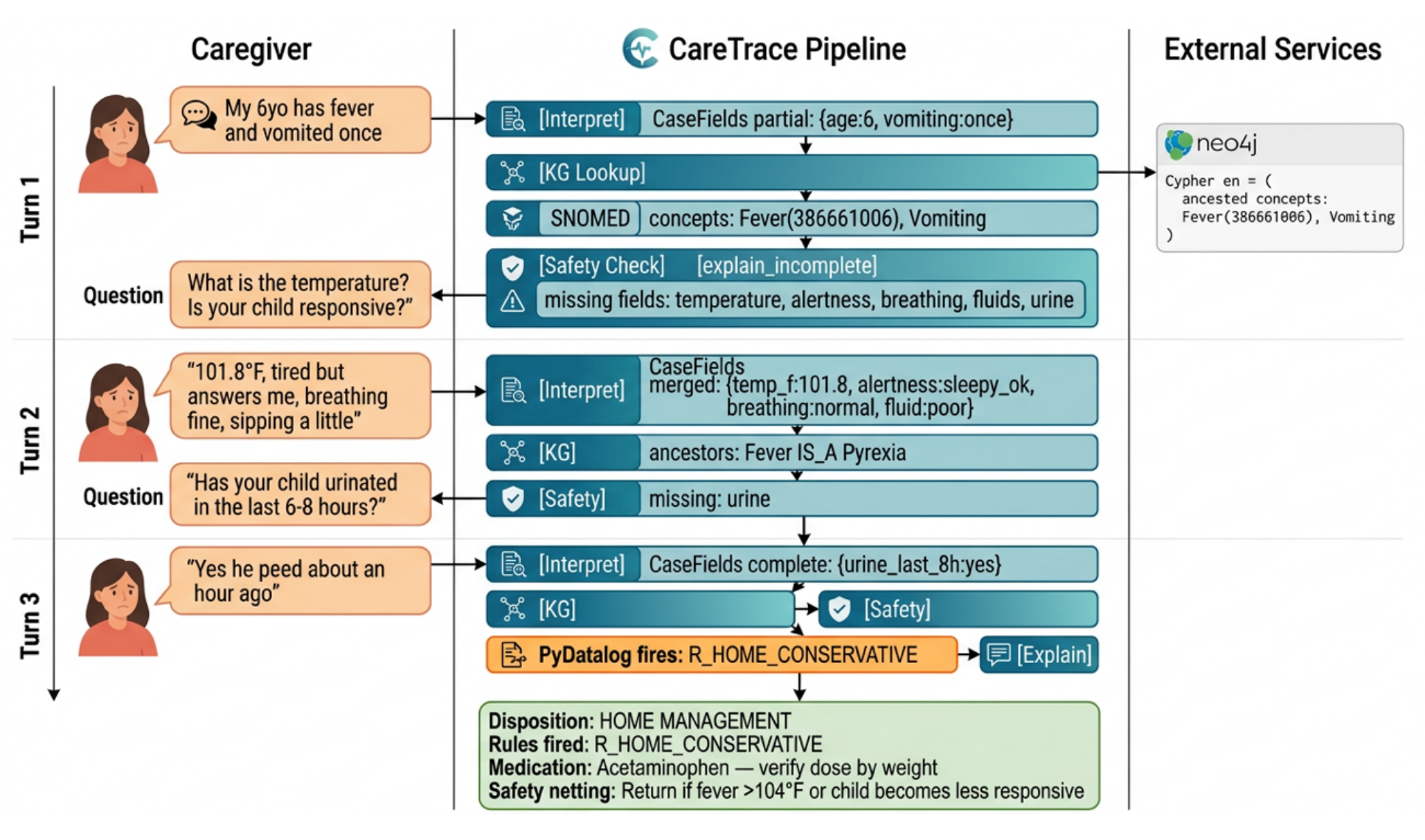
5. EVALUATION

5.1 Scenario Design

Three multi-turn scenarios cover each of the three primary dispositions. They are stored in `evaluation/scenarios.csv` as JSON arrays of caregiver utterances and replayed by `evaluation/harness.py` through the full LangGraph pipeline.

Table 3 — Evaluation scenarios

ID	Clinical summary	Expected
S1	6 yo, 101.8 °F, tired but responsive, sipping water, on amoxicillin, one emesis, urinated this evening	HOME_MANAGEMENT
S2	6 yo, 103.5 °F, barely responding, not drinking, no urine since afternoon	ER_NOW
S3	5 yo, 102.5 °F, vomited 4×, only sipping, urinated 1 hour ago	URGENT_SAME_DAY


Caregiver
CareTrace Pipeline
External Services
Turn 1
Turn 2
Turn 3
Question
Question
[Interpret]
CaseFields partial: {age:6, vomiting:once}
[KG Lookup]
SNOMED
concepts: Fever(386661006), Vomiting
[Safety Check]
[explain_incomplete]
missing fields: temperature, alertness, breathing, fluids, urine
[Interpret]
CaseFields merged: {temp_f:101.8, alertness:sleepy_ok, breathing:normal, fluid:poor}
[KG]
ancestors: Fever IS_A Pyrexia
[Safety]
missing: urine
[Interpret]
CaseFields complete: {urine_last_8h:yes}
[KG]
[Safety]
PyDatalog fires: R_HOME_CONSERVATIVE
[Explain]
neo4j

My 6yo has fever and vomited once

What is the temperature? Is your child responsive?"

"101.8°F, tired but answers me, breathing fine, sipping a little"

"Has your child urinated in the last 6-8 hours?"

"Yes he peed about an hour ago"

```
Cypher en = (
  ancested concepts:
  Fever(386661006), Vomiting
)
```

Disposition: HOME MANAGEMENT
Rules fired: R_HOME_CONSERVATIVE
Medication: Acetaminophen — verify dose by weight
Safety netting: Return if fever >104°F or child becomes less responsive

Figure 2 — Three-turn conversation flow (S1, HOME_MANAGEMENT scenario).

5.2 Extracted CaseFields Per Scenario

The table below shows the fully-accumulated `CaseFields` at the end of each scenario, illustrating how multi-turn merging builds a complete intake record:

Table 4 — Final accumulated CaseFields per scenario

Field	S1 (HOME)	S2 (ER_NOW)	S3 (URGENT)
`temp_f`	101.8	103.5	102.5
`age_years`	6	6	5
`alertness`	sleepy_ok	altered	normal
`breathing`	normal	normal	normal
`fluid_intake`	some	none	poor
`urine_last_8h`	yes	no	yes
`vomiting`	once	once	repeated
`current_meds`	[amoxicillin]	[]	[]

5.3 Symbolic Rule Trace

Table 5 — PyDatalog facts asserted and rules fired per scenario

Scenario	Key facts asserted	Rule IDs fired
S1	alertness=sleepy_ok, breathing=normal, fluid_intake=some	`R_HOME_CONSERVATIVE`
S2	alertness=altered, fluid_intake=none, urine_last_8h=no, dehydration_severe=yes	`R_ER_ALERTNESS`, `R_ER_DEHYDRATION_SEVERE`
S3	vomiting=repeated, fluid_intake=poor	`R_URGENT_REPEATED_VOMIT_POOR_FLUID`

5.4 Sample System Output (S2 — ER_NOW)

The following is the actual CareTrace reply for scenario S2 after all three turns:

```
Disposition: emergency department now.

Why (rule trace): R_ER_DEHYDRATION_SEVERE, R_ER_ALERTNESS.

Key positives/concerns:
• Reduced responsiveness, breathing, hydration, and
  urine output as hard safety signals.
• Local outbreak context (if mentioned) can raise
  suspicion for viral illness but does not override
  these gates.

What to do now:
• Go to the ER now; do not wait overnight.
• Bring temperature log, medication list, and recent
  fluid intake notes if possible.

Uncertainty: this is not a diagnosis; it is an
escalation based on risk thresholds.

(Required intake complete — exiting.)
```

The rule trace (`R_ER_DEHYDRATION_SEVERE, R_ER_ALERTNESS`) provides a full audit trail: a reviewer can identify exactly which two PyDatalog predicates fired and which `CaseFields` values triggered them, without re-running the model.

5.5 Baseline Comparison

`evaluation/baseline_llm.py` provides a vanilla `ChatOpenAI` baseline with a minimally guided system prompt and *no* symbolic scaffolding. It receives the full turn transcript and returns a free-form reply. Documented failure modes:

Table 6 — CareTrace vs. vanilla LLM baseline

Dimension	CareTrace	Vanilla LLM
CPG adherence	Guaranteed by PyDatalog rules	Best-effort; may vary by phrasing
Disposition provenance	Named `rule_ids` always present	No structured rule citation
Dosing discipline	Weight gate prevents speculative doses	May estimate doses from age alone
Paraphrase stability	Invariant once fields extracted	Disposition may shift on rephrasing
Outbreak context handling	Recorded as context; never overrides gates	May use as modifying factor on escalation
Intake completeness gate	Explicit `required_missing` checklist	No structured intake policy

5.6 Required Intake Policy

`required_missing(case)` enforces a minimum five-field intake checklist before any disposition is issued. Until all five are populated, `evaluate_triage` returns `disposition="OUT_OF_SCOPE"` with `out_of_scope_reason="incomplete_intake"`, causing the router to select `explain_incomplete` and generate targeted follow-up questions.

Table 7 — Mandatory intake fields

Field	Blocking predicate if absent
Temperature (or explicit `temp_unknown`)	`temp_f IS NULL and NOT temp_unknown`
Alertness / responsiveness	`alertness == "unknown"`
Breathing	`breathing == "unknown"`
Fluid intake	`fluid_intake == "unknown"`
Urine in last 6–8 hours	`urine_last_8h == "unknown"`

6. DESIGN DECISIONS AND TRADE-OFFS

6.1 Why PyDatalog

PyDatalog provides Python-native Datalog with backward-chaining inference and negation as failure (`~er_now(S)`). It produces enumerable rule traces via `rule_fired` predicates without separate audit infrastructure, and session-scoped facts avoid global state contamination across evaluation runs.

6.2 Why SNOMED CT via Snowstorm

SNOMED CT is the only comprehensive clinical terminology with a publicly available training server (Snowstorm at `snowstorm-training.snomedtools.org`). Using FHIR `CodeSystem/$lookup` for display term resolution ensures the KG is built from a terminologically authoritative source.

6.3 Why LangGraph

LangGraph provides a typed state machine with conditional routing, node isolation, and compile-time graph validation. Compared to a simple imperative function chain, it makes orchestration flow explicit, supports future addition of nodes without rearchitecting the pipeline, and cleanly separates node inputs and outputs.

6.4 Scope Boundaries

The following CPG criteria are not yet fully encoded and represent extension points: localized pain (neck, ears, throat distinction), rash, immunocompromised status, "3 months–2 years with fever >24 h and no other symptoms," and chronic conditions weakening the immune system.

7. RELATED WORK

Clinical decision support systems: MYCIN [1] established symbolic rule-based diagnosis. Modern CDSS like Isabel DDx and UpToDate use hybrid rule-plus-retrieval architectures. CareTrace differs in its focus on caregiver-facing conversational triage with explicit CPG provenance tracing.

LLM safety in healthcare: Singhal et al. [2] and Nori et al. [3] show GPT-4 can pass medical licensing exams but also demonstrate hallucination risk in dosing and contraindication scenarios — exactly the categories CareTrace's symbolic layer guards.

Medical knowledge graphs: Rotmensch et al. [4] built EHR-derived disease graphs; UMLS-based KGs are standard in medical NLP. CareTrace's contribution is using a CPG-text–seeded SNOMED subgraph as a runtime retrieval oracle, not just a training resource.

Neurosymbolic integration: Garcez & Lamb [5] survey neural-symbolic computing. The *interpret → symbolic → explain* pipeline pattern mirrors the neuro-symbolic concept learner (NSCL) approach: neural perception feeds symbolic reasoning which feeds interpretable output.

8. LIMITATIONS AND FUTURE WORK

Limitations:

- Evaluation is limited to three manually authored scenarios; larger-scale clinical validation is required.
- The KG is bounded to ~45 SNOMED concepts; edge cases involving rare combinations may not be grounded.
- Heuristic interpretation fails on phrasing outside its regex vocabulary.
- This is a course prototype; thresholds and escalation logic must be reviewed by licensed clinicians before any real-world deployment.

Future directions:

- Extend `CaseFields` and rules to cover unmodeled CPG bullets (rash, immunocompromise, localized pain).
- Add a `kg_rag` node that surfaces `kg_annotations` into the explanation prompt for LLM-backed responses.
- Train a fine-tuned structured-extraction model on synthetic triage transcripts.
- Multi-CPG support: RSV, croup, and urinary tract infection guidelines.

9. CONCLUSION

CareTrace demonstrates that neurosymbolic integration is both feasible and beneficial for conversational pediatric triage. By combining LLM-based natural language understanding with PyDatalog symbolic rules derived from the Seattle Children's Fever CPG, SNOMED CT knowledge graph grounding in Neo4j, and CPG-cited medication guidance, the system produces structured, auditable, and clinician-legible triage dispositions. The architecture's key properties — traceable rule firing, explicit required-intake policy, and provenance-annotated medication language — directly address the failure modes of pure generative approaches in safety-critical clinical contexts.

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Medical disclaimer: *CareTrace is a course prototype only and is not a substitute for licensed clinical decision support or professional medical advice. All dosing guidance, thresholds, and escalation logic must be reviewed and validated by licensed clinicians before any real-world deployment.*